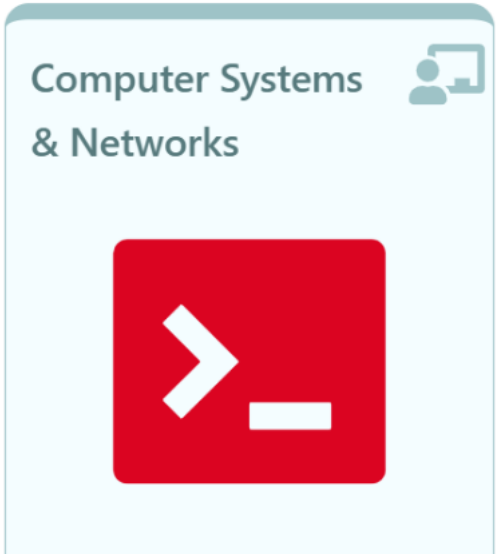
Computer Systems & Networks



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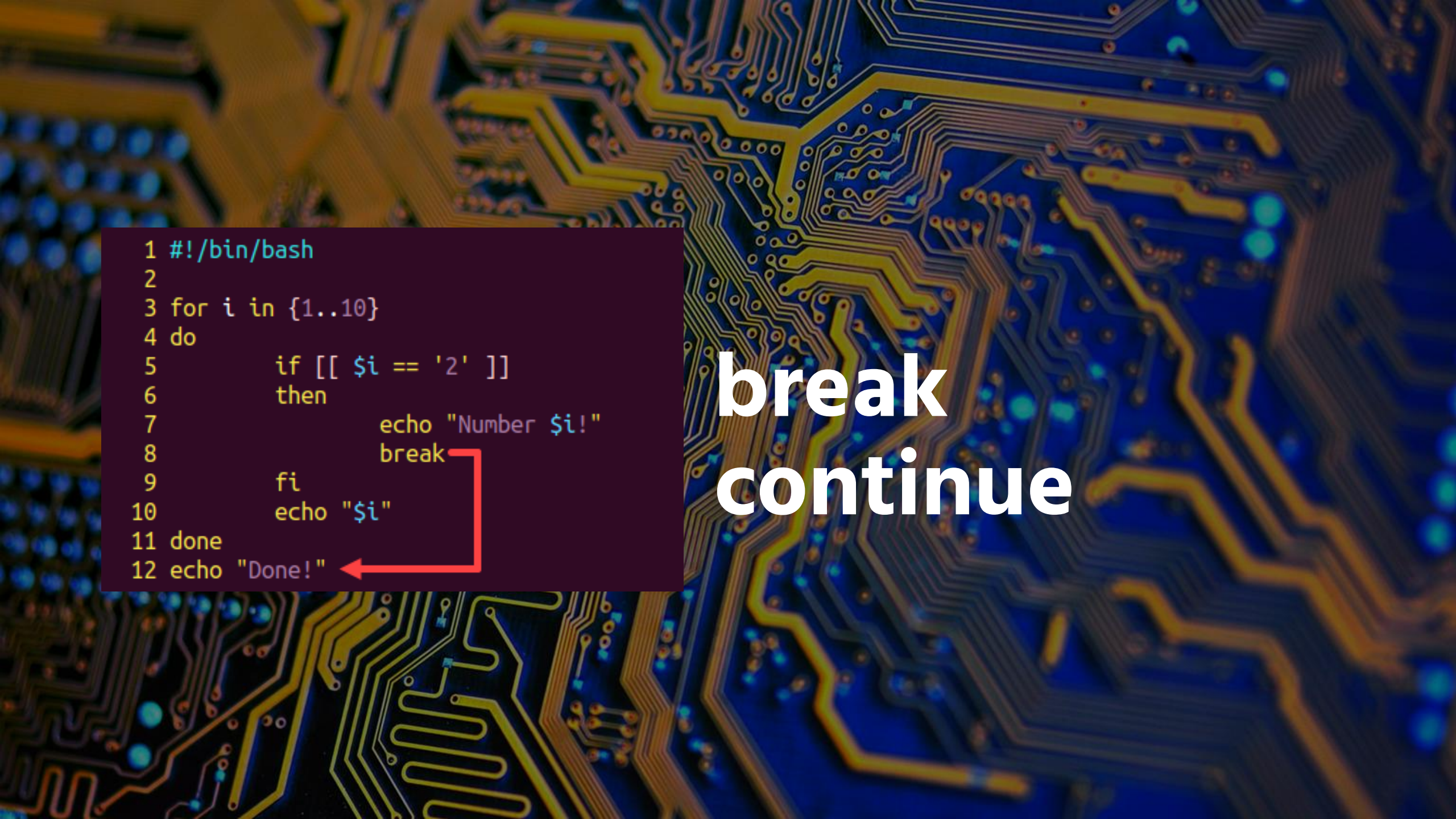

#!/bin/bash
for i in {1..10}
do
if [[$i == '2']]
then
echo "Number $i!"
break
fi
echo "$i"
done
echo "Done!"
break
continue

break statement

- used to **exit a loop early** before it naturally finishes.
- useful when you want to **stop iterating** based on a condition or event.

(1) The break statement

break

Output:
Number: 0
Number: 1
All Done!

```
i=0
while [ $i -lt 5 ]
do
        echo "Number: $i"
        ((i++))
        if [[ "$i" == '2' ]]; then
                break
        fi
done
echo 'All Done!'
```

break n usage in nested loops

The **break** statement on its own will break only **the loop that it was executed in**.

break n is used to **break out of multiple nested loops** at once

break affects only for, while, or until loops — **not if statements**.

$break n #breaks out of **n levels** of nested loops(default 1)

continue statement

continue can also be used to control the ANY loop type execution

used inside loops to **skip the rest of the current iteration** and jump to the **next iteration** of the loop

Output:
Number: 1
Skipping 2 and moving on..
Number: 3
Number: 4
All Done!

```
i=0
while [ $i -lt 5 ]
do
        ((i++))
        if [[ "$i" == '2' ]]; then
        echo “Skipping 2 and moving on..”
                continue
        fi
        echo "Number: $i"
done
echo 'All Done!'
```

used to execute different blocks of code depending on the **value of a variable or expression**

case…esac statement

Great alternative to nested if's

- used to simplify complex conditions
- starts with **case** statement and closes by **esac**

Useful when you wish to:

- create a menu in a script
- Have a user choose a menu option
- Have a reaction based on their choice

case...esac statement

- Always quote the variable: **case "$var"**
- Always include a ***** to handle unexpected input
- Patterns can be:
  - Exact: **1)**
  - OR'd: **y|Y|yes|Yes)**
  - Wildcarded: ***)**

```
read word
case “$word” in
  add)
    Statement(s) to be executed if pattern1 matches
    ;;
  remove)
    Statement(s) to be executed if pattern2 matches
    ;;
  quit)
    Statement(s) to be executed if pattern3 matches
    ;;
  *)
   Default condition to be executed
    ;;
esac
```

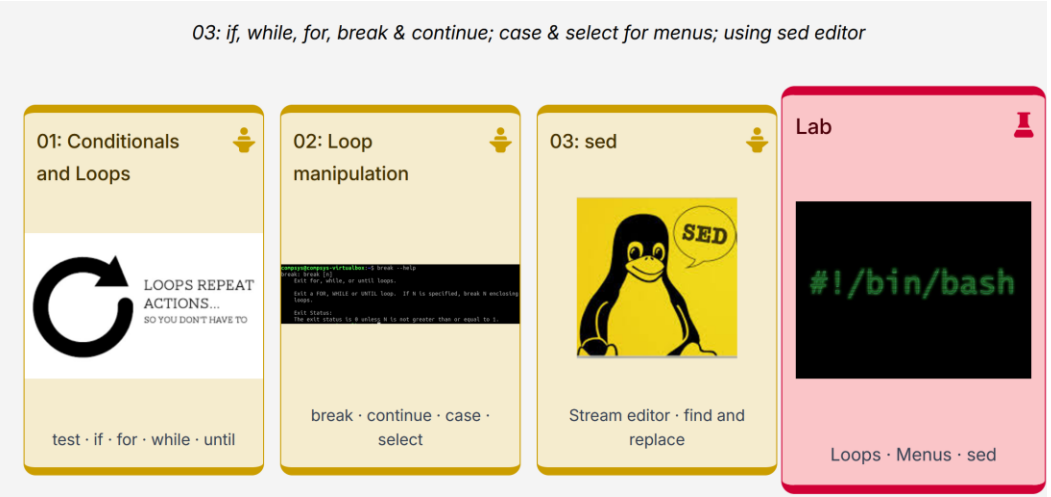
case...esac statement

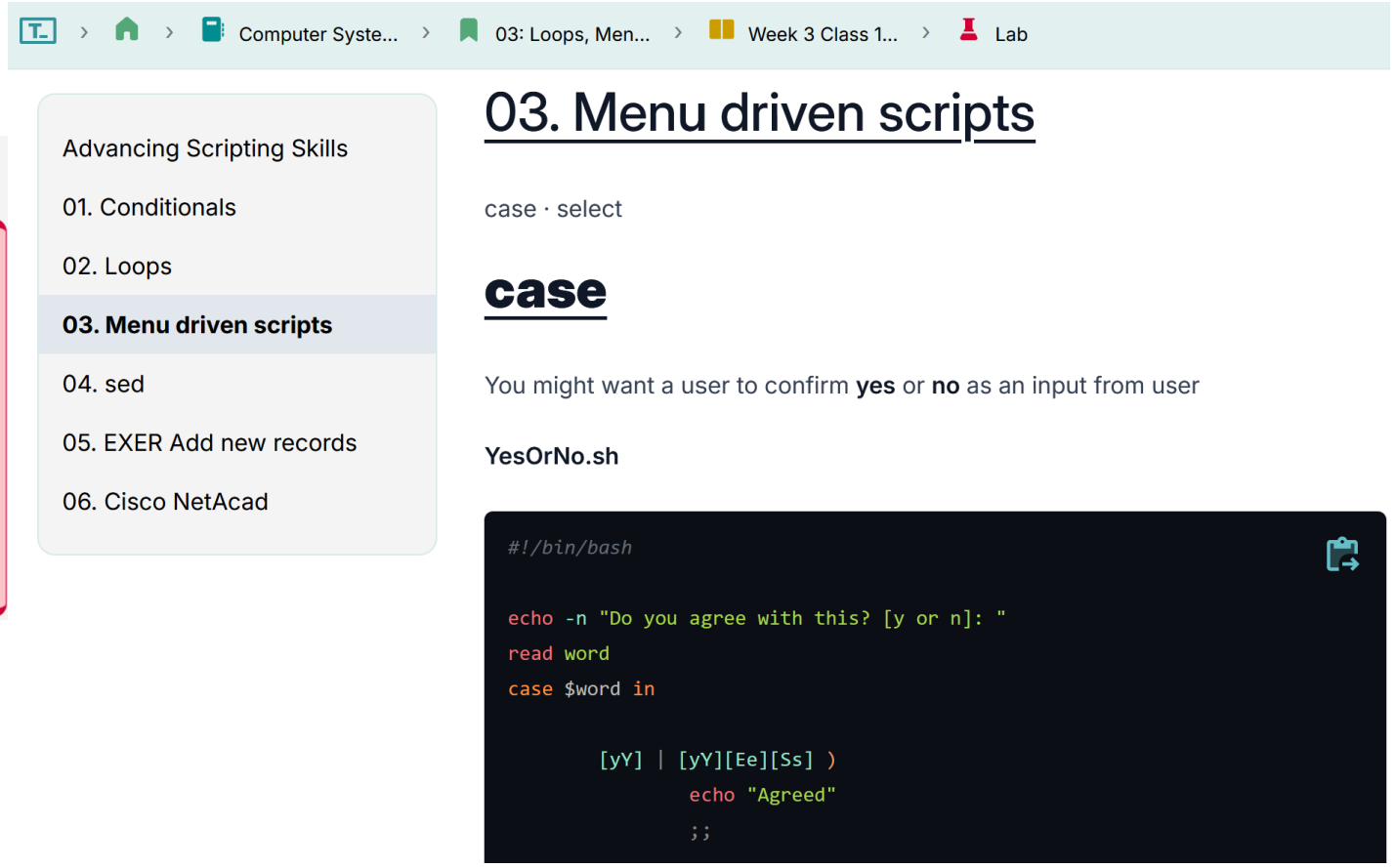
```
read userSelection
case $userSelection in
choice1)
        ACTIONS-TO-PERFORM
        ;;
choice2)
        ACTIONS-TO-PERFORM
        ;;
esac
```

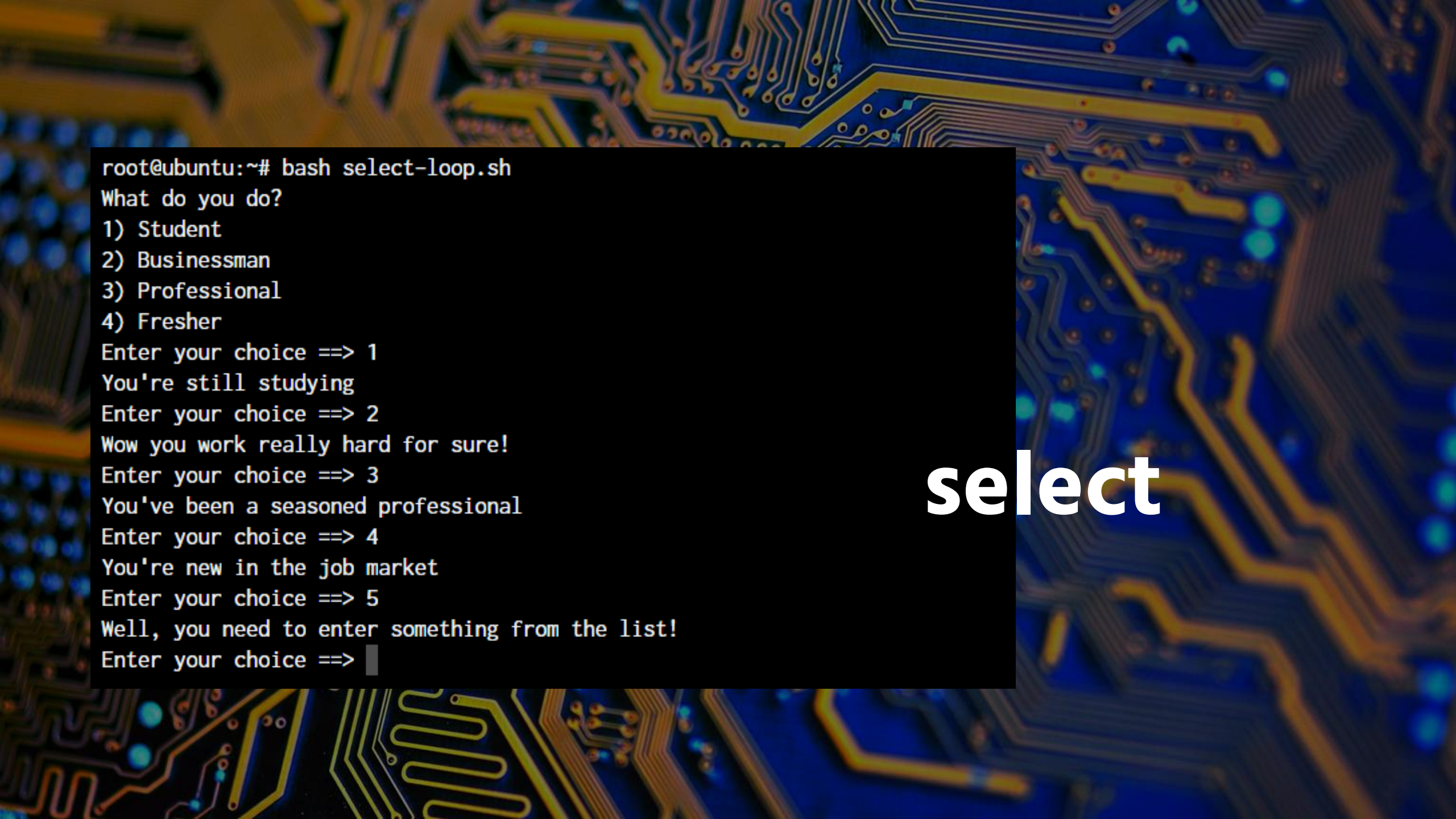
case...esac statement

```
#!/bin/bash
echo "Choose a contact to display information:"
echo "[C]hris Ramsey"
echo "[J]ames Gardner"
echo "[S]arah Snyder"
echo "[R]ose Armstrong"
read person
case "$person" in
  "C" | "c" ) echo "Chris Ramsey"
echo "cramsey@email.com"
echo "27 Railroad Dr. Bayside, NY";;
  "J" | "j" ) echo "James Gardner"
echo "jgardner@email.com"
echo "31 Green Street, Green Cove Springs, FL";;
  "S" | "s") echo "Sarah Snyder"
echo "ssnyder@email.com"
echo "8059 N. Hartford Court, Syosset, NY";;
  "R" | "r") echo "Rose Armstrong"
echo "rarmstrong@email.com"
echo "49 Woodside St., Oak Forest, IL";;
  *) echo "Contact doesn't exist.";;
esac
```

Try out this weeks Lab





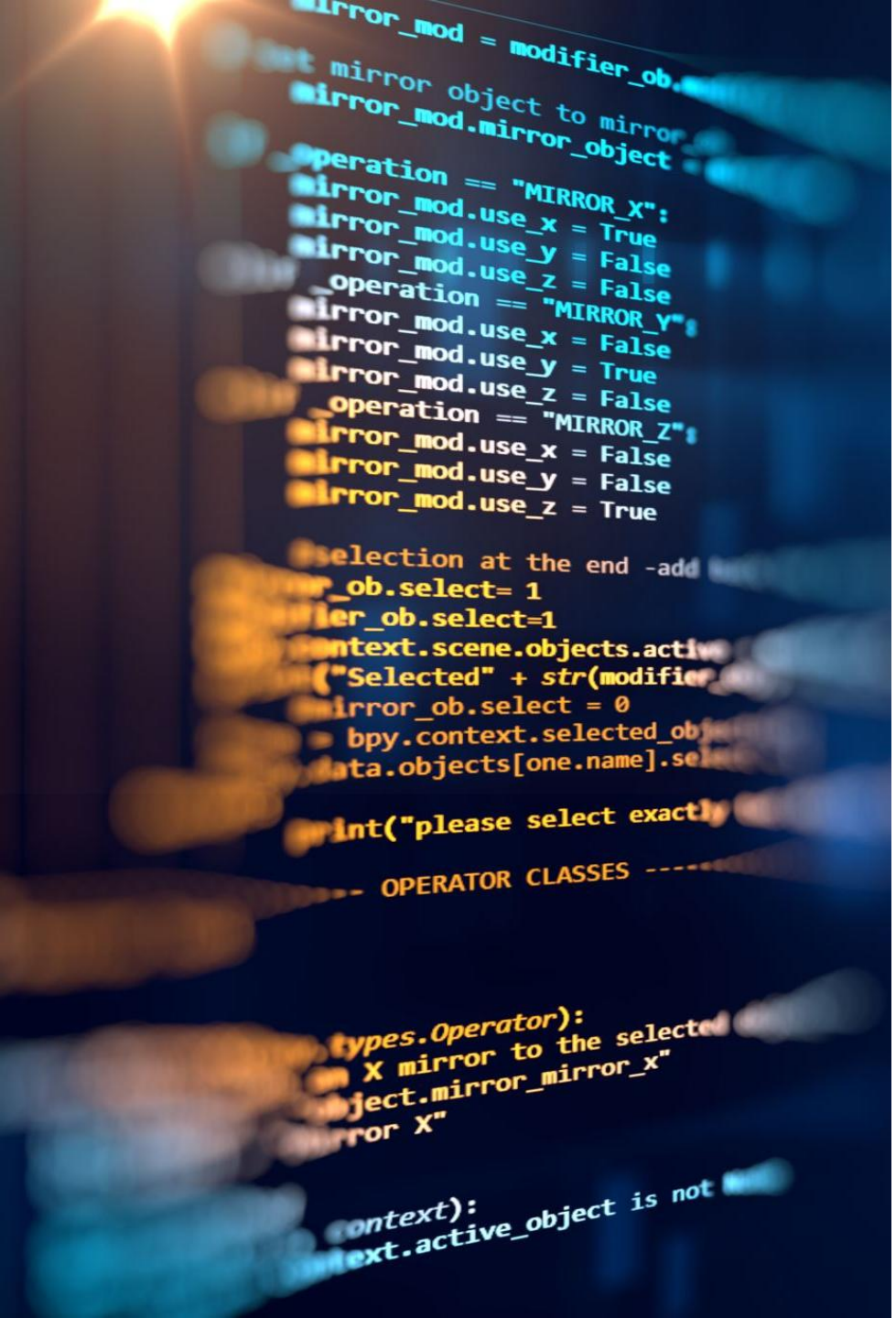

root@ubuntu:~# bash select-loop.sh
What do you do?
1) Student
2) Businessman
3) Professional
4) Fresher
Enter your choice ==> 1
You're still studying
Enter your choice ==> 2
Wow you work really hard for sure!
Enter your choice ==> 3
You've been a seasoned professional
Enter your choice ==> 4
You're new in the job market
Enter your choice ==> 5
Well, you need to enter something from the list!
Enter your choice ==>
select

select statement

- used to **create simple menus**
- Gives user a list of options to choose
- Prompt's user for an input
- **select** loops until **break** is called

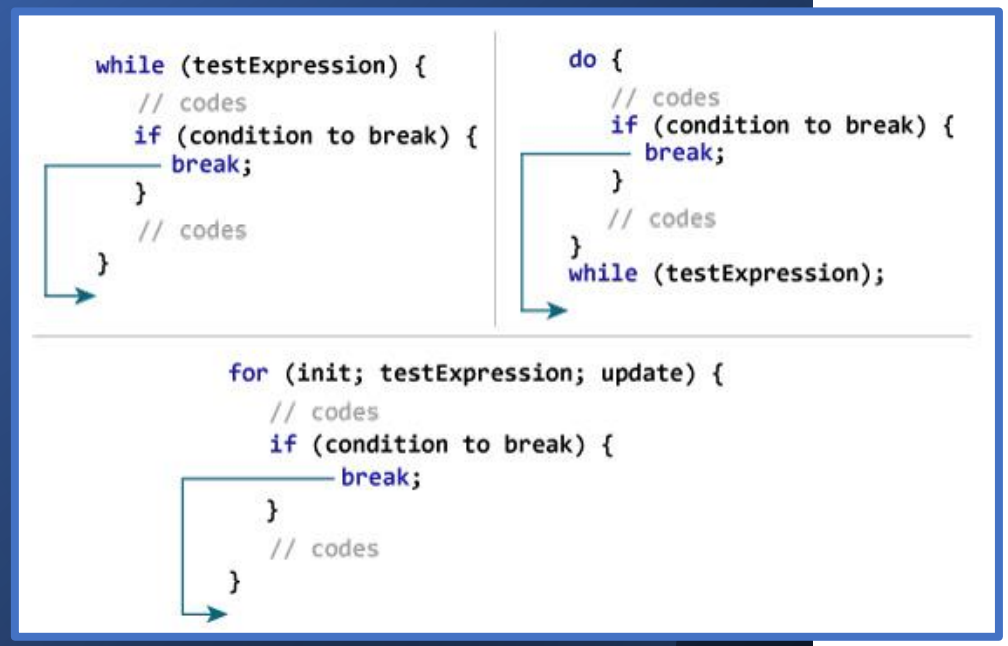
```
select variable in <list>
do
    <commands>
done
```

```
options="Add Search Quit"
PS3="Choose one option: "
select name in $options; do
        if [[ "$name" == "Quit" ]]; then
                echo "Goodbyeeeee!"
                break
        elif [[ "$name" == "" ]]]; then
                echo "Invalid option, try again."
        else
                echo "You selected: $name"
        fi
done
```



A few points with use of **select**

- No error checking is done, you must add this
- If the user hits ***enter*** without entering any data then the list of options will be displayed again.
- The loop will end when the **break** statement is issued.



Summary

while do done

- Perform a set of commands while a test is true.

until do done

- Perform a set of commands until a test is true.

for do done

- Perform a set of commands for each item in a list.

break

- Exit the currently running loop.

continue

- Stop this iteration of the loop and begin the next iteration.

select do done

- Display a simple menu system for selecting items from a list.

case

- The case statement executes any one block of commands, based on the outcome of a pattern match.

Return back to prompt Ctrl+C

Try out this weeks Lab